

Machine Learning Engineer Assessment

Freight Rate Prediction System

Prepared for: Spotter

Assessment task: Predict posted freight rates for future loads and produce the required December benchmark series.

Model pipeline: V5 — corrected and validation-controlled architecture

This report focuses on the final approach, the reasoning behind the validation design, the data-quality decisions, and the checks performed before submission. Earlier model iterations are discussed only where they explain a methodological decision.

\$130.02

3-fold CV MAE

\$129.48

Sep-Oct holdout MAE

0.8267

Holdout R^2

6.14%

CV MAPE

Evaluation note: These are internal development and out-of-time holdout results. The official score for the 12,000 validation loads is calculated by Spotter after submission.

September 2026

1. Executive Summary

The task is a supervised regression problem: estimate `posted_rate`, the freight rate in USD, from load attributes such as distance, weight, equipment, geography and date. The labeled development set contains 48,000 loads from January through October 2025. The final prediction set contains 12,000 unlabeled loads from November and December 2025, so the evaluation setting is explicitly forward-looking.

The final pipeline uses LightGBM with 33 engineered features. The main modeling decisions were driven by the structure of the data rather than by a random train/test split. Validation therefore uses expanding time windows, with a separate two-month Sep-Oct holdout to approximate the final forecast horizon.

During review, two important modeling problems were removed from earlier iterations: in-sample target encoding and a post-hoc rate multiplier. The final version uses out-of-fold target encoding with Bayesian smoothing, fits preprocessing only on the available training window, and generates December predictions directly from the model.

Key verified results

\$130.02 CV MAE	\$632.87 CV RMSE	0.8239 CV R ²	\$129.48 2-month holdout MAE
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The official structural scorer also passed: 12,000 validation predictions and 31 December predictions were accepted, and the required December chart was generated.

2. Problem and Data

The primary target is `posted_rate`. The development data contains the target; the November–December validation data does not. The separate December file fixes the physical load and varies only the date, which makes it useful for checking how the model responds to calendar changes.

File	Rows	Period	Purpose
train-test.csv	48,000	Jan 1–Oct 31, 2025	Labeled development data
validation.csv	12,000	Nov 1–Dec 31, 2025	Final unseen predictions
validation-predictions-template.csv	12,000	—	Required submission structure
december-chart-inputs.csv	31	Dec 1–31, 2025	Fixed-route benchmark

Data quality findings

- Distance is the dominant single numerical signal, with Pearson correlation of approximately 0.91 against `posted_rate`.
- 292 development rows and 145 validation rows contain negative weights. Their absolute-value distributions closely match positive weights, supporting treatment as sign-entry errors.
- Weight is missing in 300 development rows and 165 validation rows; missing `market_index` values occur in 374 and 249 rows respectively.
- There are eight cities in validation that do not appear in the development data. Geographic coordinates and smoothed location statistics are therefore important for generalization.
- Equipment classes are Dry Van, Reefer and Flatbed, with different historical rate-per-mile behavior.

Preprocessing decisions

Issue	Final treatment
Negative weight	Absolute value, supported by the matching positive/negative magnitude distributions
Missing weight	Equipment-specific median from the training window; global training median as fallback
Coordinates	Training-derived city lookup; December Lexington/Fort Wayne coordinates resolved from this lookup
Market signals	<code>market_index</code> and <code>quote_signal</code> removed from the final feature architecture
Validation leakage	All learned preprocessing parameters fitted only on the corresponding training period

3. Feature Engineering

The final model uses 33 features. They were kept deliberately compact and grouped around physical haul characteristics, geography, calendar effects and historical rate structure.

Group	Features	Purpose
Distance	distance, log_distance, distance_sq	Capture base haul cost and non-linear distance behavior
Weight	weight, log_weight	Represent payload size and its non-linear effect
Equipment	equipment_code, equip_premium	Represent equipment class and historical premium
Calendar	day_of_week, day_of_month, is_weekend, sin/cos encodings	Capture weekly and monthly timing
Coordinates	pickup/delivery lat/lon	Generalize geographically beyond city labels
Spatial geometry	haversine, circuitry, deltas, midpoint	Describe route shape and geographic orientation
Interactions	distance × weight, distance × premium, log-distance × premium	Capture combined physical effects
Historical rate	pickup/delivery encodings, equipment RPM, baseline rate	Provide smoothed historical pricing context

Why market_index and quote_signal were removed

An isolated ablation test compared the feature set with and without these two signals. In that specific comparison, CV MAE improved from \$161.07 to \$154.20 after removing them. More importantly, removing the signals made the same feature schema usable for the validation and December inputs without inventing future market values.

The larger improvement seen in the final V5 pipeline should not be attributed to this change alone: V5 also introduced out-of-fold target encoding, stricter preprocessing isolation and a revised model configuration.

4. Validation Strategy

Because the final validation period is entirely after the development period, a random split would allow the model to learn patterns from later dates while being evaluated on earlier ones. The final pipeline therefore uses expanding chronological windows.

Evaluation	Training window	Validation window	Rows
Fold 1	Jan-Jul	Aug	32,958 → 4,963
Fold 2	Jan-Aug	Sep	37,921 → 4,917
Fold 3	Jan-Sep	Oct	42,838 → 5,162
Holdout	Jan-Aug	Sep-Oct	37,921 → 10,079

Target encoding

Earlier experiments calculated route statistics on the entire development set and merged those values back into the same rows. That allows a row's own target to influence its encoded feature. The final pipeline avoids this by creating training encodings out-of-fold.

For each of five internal folds, a training row receives its encoding from the other four folds. The estimates are smoothed toward the global target mean with $m = 15$. At inference time, statistics are calculated only from the historical training window; unseen locations fall back to the smoothed global estimate.

Model configuration

Parameter	Value
Algorithm	LightGBM LGBMRegressor
Leaves / depth	63 / unlimited
Learning rate	0.03
Feature fraction	0.85
Bagging fraction / frequency	0.85 / 5
min_child_samples	20
L1 / L2 regularization	0.1 / 0.1
n_estimators	197
Random state	42

5. Model Results

The results below are from chronological validation only. They are intended to estimate out-of-time generalization; they are not Spotter's final validation score.

Evaluation	MAE	RMSE	R ²	MAPE	MedAE
Aug 2025	\$133.78	\$627.60	0.8189	6.59%	\$52.16
Sep 2025	\$126.22	\$617.54	0.8357	5.50%	\$46.89
Oct 2025	\$130.07	\$653.45	0.8172	6.32%	\$42.63
3-fold average	\$130.02	\$632.87	0.8239	6.14%	\$47.23
Sep-Oct holdout	\$129.48	\$635.27	0.8267	5.96%	\$44.84

Experiment progression

Version	Status	Main outcome
V1	Clean baseline	LightGBM baseline; CV MAE \$154.08
V2	Rejected	Deeper trees overfit; CV MAE \$180.60
V3	Rejected — leakage	Global in-sample target aggregates
V4	Rejected — heuristic	Post-hoc rate multiplier
V5	Final	OOF encoding, isolated preprocessing, direct inference

The final selection was based on the corrected validation procedure rather than on the highest score from an earlier, methodologically flawed iteration.

6. December Benchmark

The December benchmark fixes the route at Lexington → Fort Wayne, 360 miles, Dry Van and 32,000 lb, while changing the date across the 31 days of December. The final model handles this using the same feature pipeline used for the main prediction task.

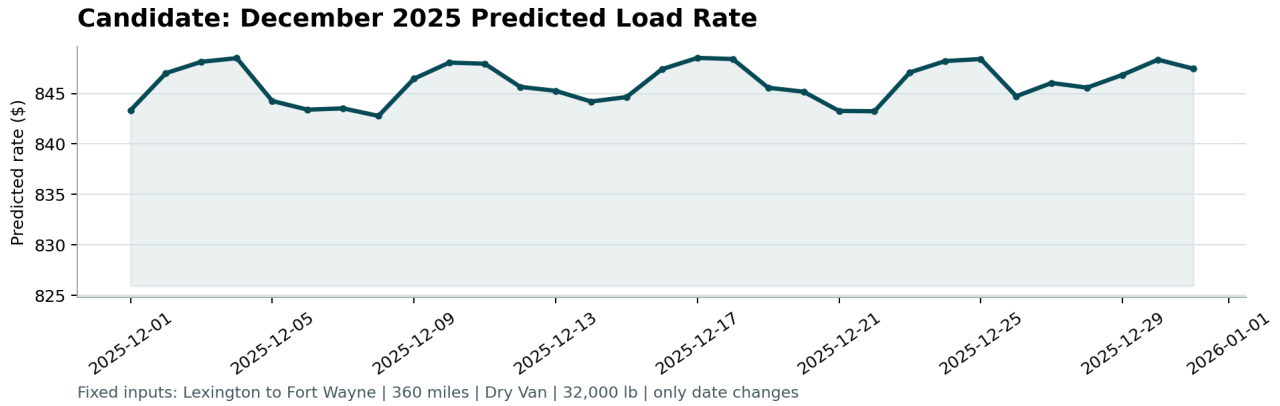


Figure 1. December 2025 predictions generated by the official scorer for the fixed benchmark corridor.

How the December inputs are handled

- Lexington and Fort Wayne coordinates are resolved from the training-derived city coordinate lookup.
- market_index and quote_signal are not carried forward from October and no artificial values are inserted.
- No external rate-per-mile multiplier is applied after prediction; the displayed values are direct model outputs.
- The model predicts a relatively narrow range, from \$842.77 to \$848.53, with a mean of \$845.98.

For reference only, the historical Dry Van mean for this lane is approximately \$857. This is context rather than an additional validation target.

7. Submission Checks and Limitations

Official structural validation

Check	Result
validation_predictions.csv	12,000 rows; IDs match validation.csv; positive finite predictions
December predictions	31 rows; all required columns retained; unique dates
Official scorer	Exit code 0
December chart	candidate_december.png generated successfully

Limitations

- The official Spotter score is unknown until the submitted predictions are evaluated against the hidden ground truth.
- Development data ends on October 31, so the model has no direct training examples for Christmas or New Year's Eve freight-market effects.
- The December route is static in distance, weight and equipment, so the tree model's temporal response is intentionally modest rather than artificially amplified.
- Historical validation results are estimates of future performance and should not be treated as guarantees.

Reproducibility

1. Install dependencies: `python -m pip install -r requirements.txt`
2. Run the pipeline: `python train_model.py`
3. Verify outputs: `python score.py --predictions validation_predictions.csv --december-predictions december_predictions.csv`

Conclusion

The final approach combines a time-aware validation design with compact tabular feature engineering and a regularized LightGBM model. The main engineering focus was making preprocessing and target encoding behave the same way they would in a future-prediction setting. The resulting pipeline passed the supplied structural scorer and produced the required validation and December prediction files.

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